



Improving Grape Quality

Lesson Aim

Explain different ways to ensure or improve grape quality.

Improving Grape Quality

Improving grape quality is a major aspect of viticulture operations. High quality vines not only provide larger yields, they also have desirable characteristics such as improved berry uniformity, increased berry weight, and reduced post-harvest shatter.

A variety of methods are used to improve grape quality, and all stages of crop growth are manipulated to promote the development of quality grapes, from planting through to harvest and post-harvest treatment of the grape.

Plant Stock

The use of appropriate plant stock can have a significant impact on grape quality. Growing varieties that are appropriate to the locality makes the viticulturist's task much easier, since growth characteristics can be more easily manipulated when the plant is suited to the area. When selecting stock, both the characteristics of the rootstock and the scion are important – they should be suited to the area and compatible with each other.

Crop Management

Once vineyards are established, quality is most frequently improved through effective management of irrigation, soil fertility, and vine training and pruning. Various aspects of crop management can be utilised to improve grape quality:

- Irrigation can be timed to put the plant under water stress at key times during fruit development. Some viticulturists suggest that this can improve the quality of grapes for winemaking.
- Careful management of fertility ensures a good balance between vine growth and fruit development. Insufficient fertility can lead to poor vine growth, which can in turn stunt growth and development of fruit. Excessive fertility can lead to rampant vine growth, with accompanying reduction in both fruit quality and quantity.
- Pruning and training of vines is frequently utilised to improve grape quality. Vegetative growth is pruned to allow more sunlight into, and to minimise humidity within, the crowns of grape vines. Inflorescences are also pruned to improve the development of grape bunches.

Special Techniques

Special techniques such as girdling and pruning the tops of the vine just prior to cap fall, and use of chemicals such as growth retardants, can improve yields and quality of produce. Increasingly other special techniques such as chemical growth regulation and gene technology are also being used to promote the development of quality grapes.

Post-Harvest Impact on Quality

The post-harvest treatment of grapes also has a significant impact on quality. This can include storage under appropriate conditions for table grapes, controlled drying conditions for sultana production, and the addition of products in wine production.

Careful handling of crops during and after harvest will prevent damage to fruit and can improve the marketability of the harvest. Careful handling may involve developing harvest techniques to reduce stress and bruising to the fruit, cold storage or humidity-controlled storage to prevent fruit spoilage, and good packaging methods and materials. Various methods and inputs are used in the wine industry to maintain product consistency. Products such as sugar or chemicals like sulphur dioxide may be added to wine to stabilise it.

Techniques To Improve Grape Quality

Improving Flower and Fruit Set

Grapes have bisexual flowers – both the male and female flower parts occur on the same organ. The inconspicuous flowers of grapes are held in racemes (bunches) which, after fertilisation, become the seed-bearing fruit. The fruit/seed is not considered to be either male or female – it is a self-contained organism capable of becoming a new plant.

At the start of spring, the leaf buds burst open to produce a vegetative shoot well before the flower opens (up to eight weeks). When the flower buds do burst, the flowers need to be pollinated for good fruit set. As most varieties are self-pollinated, the process is assisted by light winds rather than insects. Warm dry weather will result in better pollination than cold, wet weather. Flowers appear two or three days later. During flowering the soil must not be allowed to dry out as water stress during this period reduces fruit set.

Unfertilised flowers usually drop off quickly. Some unfertilised flowers do form berries but these are usually small and prone to falling from the vine. Occasionally, and on certain varieties, decent berries may develop from unfertilised flowers.

There are three ways to assist better fruit set:

- Prune the tips of growing shoots a few days before flowering. Doing this removes the competition for nutrients between the flowers and rapidly growing tips of shoots. The theory is that temporarily removing the vegetative competition for nutrients and water will provide a better supply to the flowers during this critical period.
- Girdling - see below.
- Applying a chemical growth retardant. By reducing shoot growth, fruit set is increased. This works on light-yielding varieties but it is not suggested for varieties that do not have a fruit set problem. The chemical most commonly used, Chlormequat (also called CCC or Cycocel), is sprayed at 500-700ppm active ingredient.

Second Set

It is common for some grape varieties to produce a second bunch of flowers later in the season, on laterals. This second set is unlikely to ripen in cooler climates. Treatment with a chemical growth retardant can promote the development of a second set.

The second set can be a problem for mechanically harvested grapes because these unripened, developing grapes tend to be harvested at the same time as the ripe, earlier set grapes. When crushed, they reduce the overall quality of the juice (e.g. resulting in lower sugar levels).

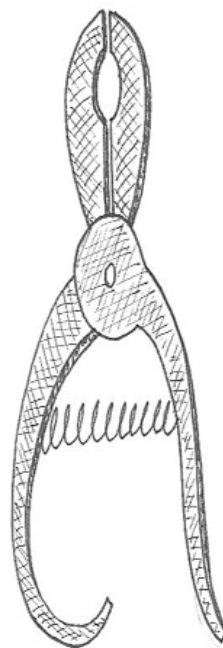
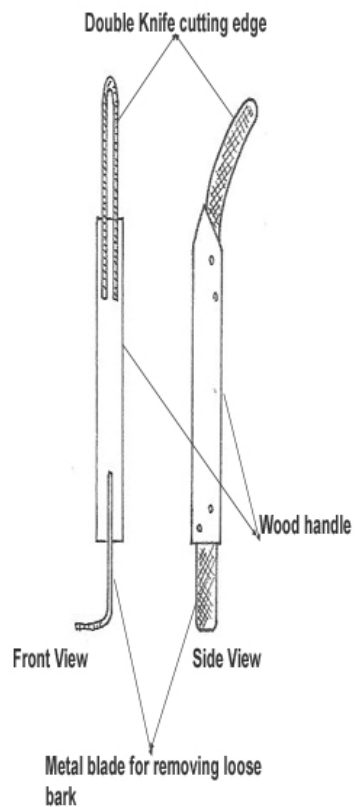
Girdling

Girdling involves cutting a section of bark away from the trunk or stem. The process puts stress on the plant, causing food materials and hormones to collect above the cut, leading to a better quality grape.

The process involves making two parallel cuts around 4 mm apart and removal of some bark. The wood should not be damaged.

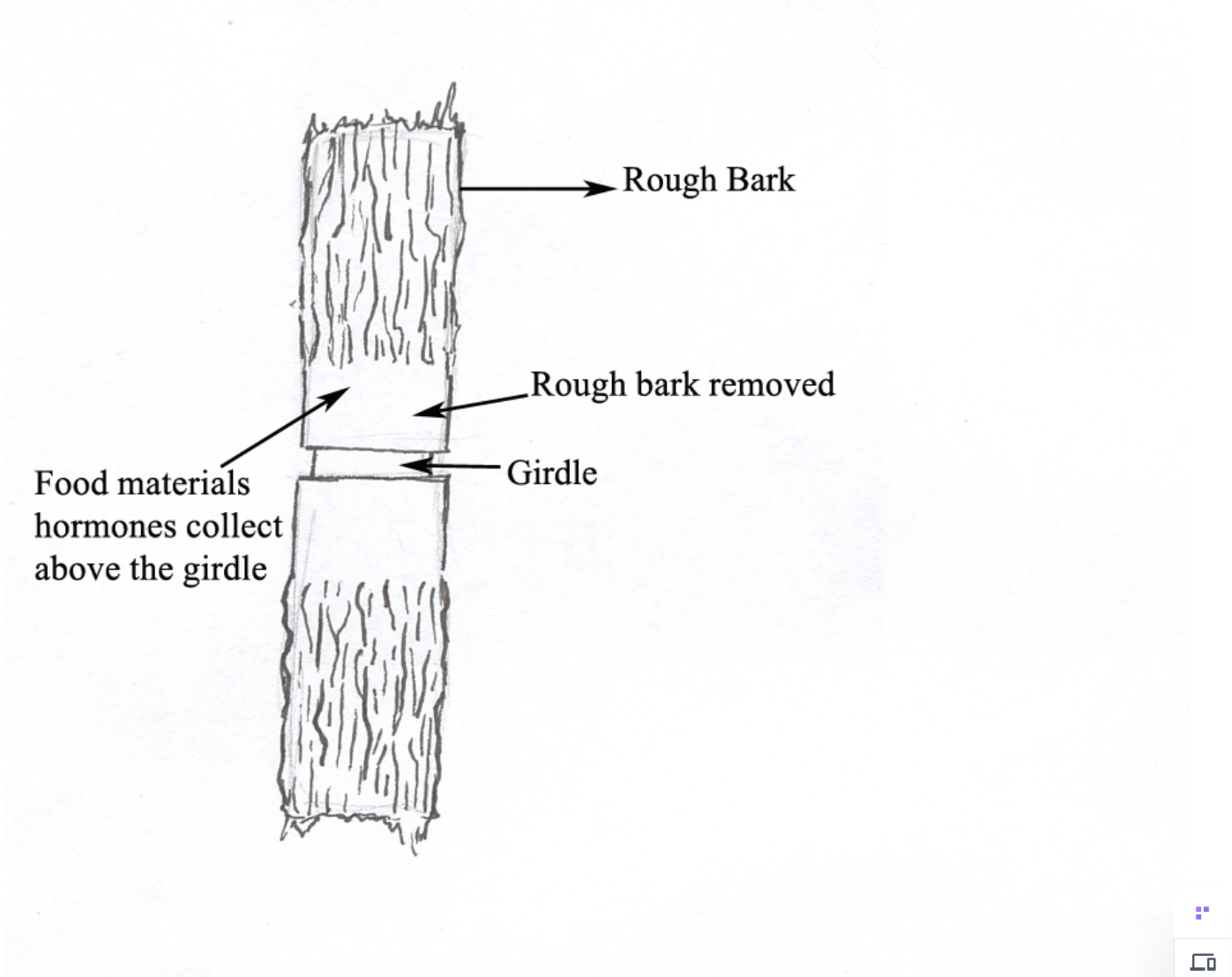
Girdling is done using special girdling tools: either a scissor- or pliers-like tool with two blades which come together to circle the stem, or a special knife with a curved, sharp cutting blade at one end and a blunt metal blade at the other end (for removing the rough outer bark before making the cut).

Girdling is done around a week before cap fall (i.e. when the cap falls from the top of the flower bud; before the flower bursts open). Do not use this on anything but varieties with poorer fruit set. Properly done, girdling can increase yield by up to 100%.



Cane girdling tools

Tools used for girdling grape vines



Trunk girdling

Berry Thinning

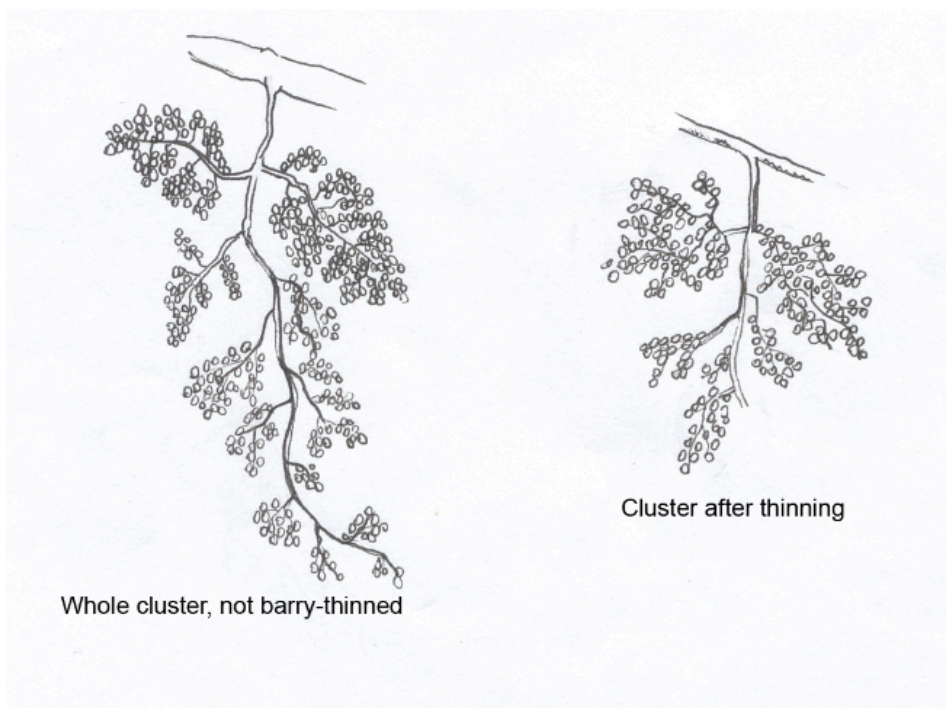
By hand

The young bunches of grapes may have parts cut off them with a pair of secateurs. This should be done as soon as possible after the fruit has set, and the young berries can be seen. Cutting off 20-40% of the berries will usually cause the remaining berries to grow larger and be of better quality.

Chemically

Berries can be thinned using a chemical spray (e.g. GA_3). Depending on the concentration of chemical sprayed, and the stage of crop growth when it is applied, different results can be achieved.

Berry thinning



Berry thinning

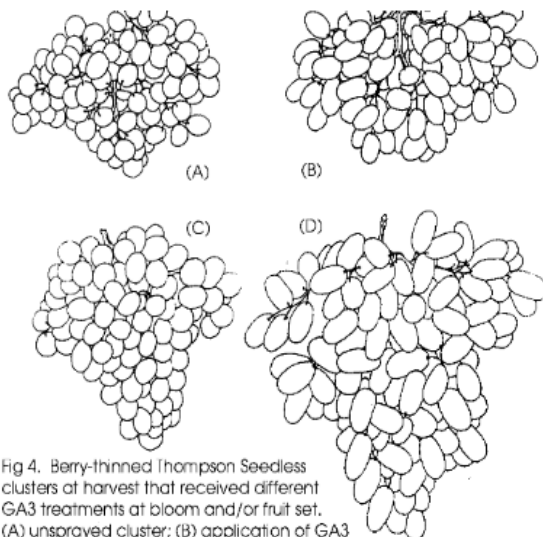


Fig 4. Berry-thinned Thompson Seedless clusters at harvest that received different GA3 treatments at bloom and/or fruit set. (A) unsprayed cluster; (B) application of GA3 at 10ppm at bloom; (C) application to GA3 at 40ppm at fruit-set; (D) application of GA3 at 10ppm at bloom, and at 40ppm at fruit-set. Sprays at bloom loosen the clusters: repeat spray results in the largest berries.



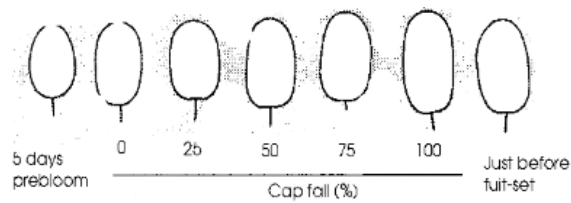


Fig 5. Effect of stage of flower and or berry development at time of treatment with GA3 on shape and size of mature Thompson Seedless berries at harvest.

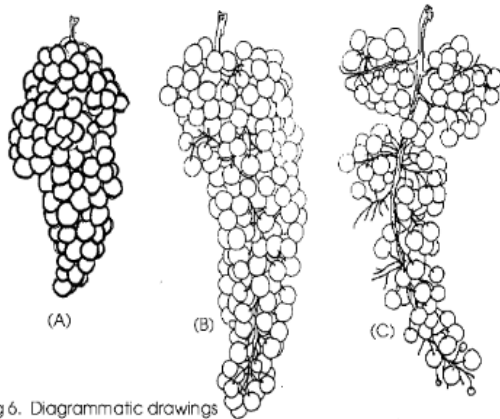


Fig 6. Diagrammatic drawings of Zinfandel, a compact-clustered wine grape, sprayed two weeks before bloom with GA3 at 0ppm (A), 7.5ppm (B), and 20ppm (C). Satisfactory loosening of cluster resulted with GA3 at 7.5ppm (B), but 20ppm (C) reduces crop.

Chemically-thinned grape clusters

Chemically-thinned grape clusters

